

Cherry-pick

Sometimes you want one commit from another branch, not the whole merge

Find the commit, then pick it

- On the source branch, use log to find the short hash you need
- Check out the target branch, then cherry-pick that hash
- Git replays the diff; if it applies cleanly, you get a fresh commit
- If not, resolve conflicts, stage, and cherry-pick continue, or abort to undo the attempt

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Cherry-pick lab

Copy one commit onto another branch — clean picks vs conflict cases — then continue or abort.

Starter example: On release/0.1 (frozen alu). Cherry-pick the docs commit from main cleanly, or try the mul commit to hit a conflict.

Load starter example

Challenges 0 / 22 cleared

Quiz: what: Cherry-pick applies how many commits by default? Answer: 1

Answer

Show hintCheckNextIdle

Quiz: whatQuiz: SHAQuiz: useOn releaseSelect docsClean pickConflict pickResolve + continueAbort

Quiz: continueQuiz: abortCheckout mainQuiz: vs mergeAlready ancestorNew SHA differsQuiz: merges

Continue blockedDocs textQuiz: messageFreeze remainsBoth outcomesQuiz: hotfix flow

Idea

Cherry-pick

Apply one commit's patch onto HEAD as a *new* commit (new SHA).

Conflicts

If the same lines diverged, resolve, then --continue or --abort.

Starter on releaseSelect mul (conflict)Select docs (clean)

Cherry-pick lab starter

Real Git practice

wsl - module12-git-cherry-pick-lab/examples/cherry_pick

```
# Track A - real Unix
# real shell session (Track A)

# practice repo: /tmp/git-cp-5im2B8

$ git init

$ echo "base" > notes.md

$ git add notes.md

$ git commit -m "Base"

$ git branch -M main

$ git checkout -b feature/fix

$ echo "hotfix" > fix.txt

$ git add fix.txt

$ git commit -m "Add hotfix on feature"

$ git log --oneline feature/fix
ef68f33 Add hotfix on feature
1150ed0 Base

$ git checkout main

$ git cherry-pick ef68f33
[main ef68f33] Add hotfix on feature
Date: Mon Jul 20 14:32:00 2026 +0800
1 file changed, 1 insertion(+)
create mode 100644 fix.txt

$ git log --oneline --all
ef68f33 Add hotfix on feature
1150ed0 Base
```

Real shell — log, cherry-pick one commit onto main

Real Git practice — try these

```
# git init — create a new repository here  
git init
```

```
# echo ... > notes.md — base commit on main  
echo "base" > notes.md
```

```
git add notes.md
```

```
git commit -m "Base"
```

```
git branch -M main
```

```
# git checkout -b feature/fix — branch for the commit to copy  
git checkout -b feature/fix
```

```
echo "hotfix" > fix.txt
```

Pitfalls to watch

- Cherry-pick creates a duplicate commit
- Avoid picking merge commits unless you know parent selection
- If conflicts appear, finish with continue or abort, do not leave a half-finished pick
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and complete a clean pick and a conflict recovery
- On real Git, cherry-pick one commit from a feature branch onto main
- When you are ready